

1. What is Artificial Intelligence (AI)?

A: AI is the ability of machines to perform tasks that normally require **human intelligence**, such as learning and decision-making.

2. What is Machine Learning?

A: Machine Learning is a subset of AI where systems learn patterns from **data** without being explicitly programmed.

3. What is Python commonly used for?

A: Python is used for **AI, Data Analysis, Automation**, Web Development, and scripting because it is simple and powerful.

4. What is OOP?

A: Object-Oriented Programming (OOP) is a programming style that organizes code into **Classes** and **Objects**.

5. What are the four pillars of OOP?

A: The four pillars are **Encapsulation, Inheritance, Polymorphism**, and **Abstraction**.

6. What is a Database?

A: A **Database** is a structured collection of data that can be stored, managed, and retrieved efficiently.

7. What is SQL?

A: SQL (Structured Query Language) is used to **Create, Read, Update**, and **Delete** data in databases.

8. What is the difference between DELETE and DROP?

A: DELETE removes data from a table, while **DROP** removes the entire table structure and data.

9. What is a Primary Key?

A: A Primary Key uniquely identifies each record in a table and cannot contain duplicate values.

10. What is HTML?

A: HTML defines the structure and content of a web page using elements and tags.

11. What is CSS?

A: CSS is used to control the appearance, layout, and styling of web pages.

12. What is UI and UX?

A: UI (User Interface) is how a product looks, while **UX (User Experience)** is how users interact with and feel about the product.

13. Why is Software Testing important?

A: Software Testing helps find bugs, improve quality, and ensure the software works as expected.

14. What is Agile?

A: Agile is a software development methodology that focuses on **Iteration, Collaboration,** and continuous improvement.

15. What is Scrum?

A: Scrum is an Agile framework that uses short development cycles called **Sprints**.

16. What is Software Architecture?

A: Software Architecture is the high-level design that defines how system components interact with each other.

17. What is a REST API?

A: A REST API allows applications to communicate using standard **HTTP methods** like GET, POST, PUT, and DELETE.

18. What is the MVC Pattern?

A: MVC stands for **Model**, **View**, and **Controller**, which separates data, user interface, and business logic.

19. What is the difference between Android and iOS development?

A: Android typically uses Kotlin or Java, while **iOS** uses Swift; both are used to build mobile applications.

20. What is Version Control and why is it important?

A: Version Control (such as **Git**) tracks code changes, supports collaboration, and allows rollback to previous versions.

1. Tell me about yourself.

Answer:

I am Mohamed Fathalla. I am recently graduated from Computer Engineering department at Menoufia University. I am interested in AI, Data Engineering, and Software Development. I have studied Python, SQL, OOP, and database fundamentals through university courses and training programs. Recently, I joined training programs in AI and Linux administration to improve my technical skills. My goal is to gain practical experience and start my career as a software and AI engineer.

Keywords: Computer Engineering, Python, SQL, AI, Software Development, Training Programs

2. Why did you choose this track?

Answer (AI Track):

I chose this track because AI is one of the fastest-growing fields. I enjoy working with data and solving problems using technology. I want to build a strong foundation in AI and gain practical skills that can help me in real projects.

Keywords: AI, Data, Problem Solving, Practical Skills

3. Why should we select you?

Answer:

I am eager متحمس to learn, I have a solid understanding of the fundamentals, and I am committed to improving my technical skills. I also have experience working on academic projects and learning independently.

Keywords: Fast Learner, Fundamentals, Projects, Self-Learning

4. What are your strengths?

Answer:

My strengths are **problem solving, teamwork**, and my willingness to learn new technologies quickly.

Keywords: Problem Solving, Teamwork, Learning

5. What is your biggest weakness?

Answer:

My English communication skills are still improving, but I practice daily by reading technical documentation and watching technical content.

Keywords: Improvement, Technical Documentation, Learning

6. Why are you interested in AI?

Answer:

AI can automate tasks, analyze large amounts of data, and help solve real-world problems. I find this combination very interesting.

Keywords: Automation, Data Analysis, Real-World Problems

أسئلة تكنولوجية سهلة جدًا تتكرر كثيرًا

What is the difference between AI and Machine Learning?

AI is the broader field. **Machine Learning** is a subset of AI that learns from data.

What is Python?

Python is a high-level programming language known for its simplicity and strong support for AI and automation.

What is SQL used for?

SQL is used to manage and query data stored in databases.

What is OOP?

OOP is a programming paradigm based on **Classes** and **Objects**.

What is a REST API?

A REST API allows different applications to communicate using **HTTP requests**.

What is Agile?

Agile is a development methodology focused on **iteration**, **collaboration**, and continuous improvement.

Software Development و AI شيوعًا في تدريبات Screening احفظ دول كويس جدًا لأنهم أكثر أسئلة

Why did you choose this program?

I chose this program because it combines both **Artificial Intelligence** and **Software Engineering**. I am interested in understanding how AI can be integrated into the software development lifecycle. I also want to learn cloud technologies, DevOps practices, and AI-native development methodologies that are highly demanded in the industry today.

Keywords: AI, Software Engineering, Cloud Computing, DevOps, AI-Native Development

What do you know about the program? لو سألوك

The program focuses on building software in the AI era. It teaches AI fundamentals, Generative AI, Cloud Computing, Big Data, AI-Native Software Engineering, Adaptive Architecture, and DevOps. It also emphasizes collaboration with AI tools and agents during software development.

Keywords: Generative AI, Cloud Computing, Big Data, DevOps, AI Agents

What is Generative AI? لو سألوك

Generative AI is a type of AI that can create new content such as text, images, code, and audio based on patterns learned from data.

Keywords: Generate Content, Text, Images, Code

What is Prompt Engineering? لو سألوك

Prompt Engineering is the process of designing effective prompts to get accurate and useful responses from AI models.

Keywords: Prompts, AI Models, Better Results

لؤ سألؤك: What is Cloud Computing?

Cloud Computing means accessing computing resources such as servers, storage, and databases over the internet instead of managing them locally.

Keywords: Servers, Storage, Internet

لؤ سألؤك: What is DevOps?

DevOps is a culture and set of practices that improve collaboration between development and operations teams and automate software delivery.

Keywords: Automation, Collaboration, CI/CD

لؤ سألؤك: What is an API?

An API allows different applications to communicate and exchange data with each other.

Keywords: Communication, Data Exchange

لؤ سألؤك: What is Machine Learning?

Machine Learning is a branch of AI where systems learn from data and improve their performance without explicit programming.

Keywords: Data, Learning, Prediction

لؤ سألؤك: What is the difference between AI and Machine Learning?

AI is the broader field of creating intelligent systems, while Machine Learning is a technique used to achieve AI by learning from data.

Keywords: AI, Machine Learning, Data

لؤ سألؤك: Do you have practical experience?

جأؤب بصرأحة لكن بشكل إيجابى

I am still building my practical experience. I have completed academic projects and training programs in AI, Linux, and Data Engineering. I am eager to learn and apply my knowledge in real-world projects.

Keywords: Projects, Training Programs, Learning

أهم نقطة

DEPI في SQL و Data Engineering ، وتدرس NTI في Linux Red Hat أنت طالب هندسة حاسبات، دخلت تدريب ، فاذكرها عندما تعرف بنفسك. لا تدّع خبرة عملية غير موجودة، وركز Screening Interview هذه نقاط قوية جدًا في **learning ability** و **technical foundations** على.

Tell me about yourself

I am Mohamed Fathalla, and I am recently graduated from Computer Engineering department of engineering faculty. I have a strong interest in Artificial Intelligence, Software Engineering, and Data-related technologies. During my studies, I learned programming fundamentals, Object-Oriented Programming, databases, SQL, and software development concepts. I also joined technical training programs in Linux Administration, AI, and Data Engineering to strengthen my practical skills. I enjoy learning new technologies and understanding how software systems are designed and built.

Keywords: Computer Engineering, Software Engineering, AI, SQL, Linux, Data Engineering

Why did you choose this program?

I chose this program because it combines two areas that I am highly interested in: Artificial Intelligence and Software Engineering. Most programs focus only on AI models or only on software development, but this program focuses on how AI can become part of the entire software development lifecycle. I am especially interested in learning AI-native development, cloud computing, DevOps, and adaptive software architecture. I believe these skills will be very important for the future of software engineering, and I want to be prepared for that future.

Keywords: AI-Native Development, Software Engineering, Cloud Computing, DevOps, Software Architecture

Why should we accept you?

I believe I am a strong candidate because I have a solid technical foundation and a strong desire to learn. Throughout my studies and training programs, I have continuously worked on improving my skills in programming, databases, Linux, and AI-related topics. I understand that I still have a lot to learn, but I am committed to continuous improvement and I enjoy challenging myself with new technologies. I believe I can benefit greatly from this program and contribute positively to the learning environment.

Keywords: Technical Foundation, Continuous Learning, Commitment, Problem Solving

What do you know about this program?

From my understanding, this program focuses on software development in the AI era. It covers Artificial Intelligence, Machine Learning, Generative AI, Cloud Computing, Big Data, AI-Native Software Engineering, and Adaptive Software Architecture. One thing that makes this program unique is that it treats AI not only as a tool but also as an active participant in the development process. It also emphasizes ethical AI usage, software automation, and collaboration with AI systems throughout the software lifecycle.

Keywords: Generative AI, Cloud Computing, Big Data, AI-Native Software Engineering, Ethics

What is Artificial Intelligence?

Artificial Intelligence is the field of creating systems that can perform tasks that normally require human intelligence. These tasks may include learning from data, recognizing patterns, making decisions, understanding language, and solving problems. AI is used today in many applications such as recommendation systems, chatbots, computer vision, and autonomous systems.

Keywords: Human Intelligence, Learning, Decision Making, Pattern Recognition

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that allows systems to learn from data and improve their performance without being explicitly programmed for every task. Instead of writing specific rules, we provide data, and the model identifies patterns and relationships that can be used for predictions or decision-making.

Keywords: Data, Patterns, Predictions, Learning

What is Generative AI?

Generative AI is a type of Artificial Intelligence that can create new content based on patterns learned from existing data. It can generate text, images, videos, audio, and even software code. Examples include AI chatbots, image generation systems, and coding assistants.

Keywords: Generate Content, Text, Images, Code

What is Prompt Engineering?

Prompt Engineering is the process of designing and refining prompts to obtain better responses from AI systems. A well-structured prompt helps the model understand the task clearly and produce more accurate and useful results. It is becoming an important skill when working with modern AI applications.

Keywords: Prompts, AI Systems, Accurate Results

What is Cloud Computing?

Cloud Computing is the delivery of computing services such as servers, storage, databases, and networking over the internet. Instead of buying and maintaining physical hardware, organizations can use cloud platforms to access resources when needed. This provides scalability, flexibility, and cost efficiency.

Keywords: Servers, Storage, Scalability, Flexibility

What is DevOps?

DevOps is a set of practices that combines software development and IT operations to improve collaboration and automate software delivery processes. The goal is to build, test, and deploy applications more efficiently and reliably. Automation and continuous integration are important parts of DevOps.

Keywords: Automation, Collaboration, CI/CD, Deployment

What is Software Architecture?

Software Architecture is the high-level structure of a software system. It defines how different components interact with each other and how the system is organized. Good architecture improves scalability, maintainability, performance, and reliability.

Keywords: System Design, Scalability, Maintainability, Reliability

What are your future goals?

My short-term goal is to strengthen my skills in AI, software engineering, cloud technologies, and modern development practices. My long-term goal is to become a software engineer who can design and build intelligent systems that solve real-world problems. I also want to continue learning and adapting to emerging technologies in the AI field.

Keywords: AI, Software Engineering, Cloud Technologies, Continuous Learning

OOP

Q: What is the difference between a Class and an Object in OOP?

Answer:

A **Class** is a blueprint or template that defines the attributes and behaviors of an object. An **Object** is an actual instance created from that class.

Example:

If **Car** is a class, then a specific car like "My Toyota" is an object.

Keywords: Class, Blueprint, Object, Instance

Q: State the four pillars of OOP.

Answer:

The four pillars of **Object-Oriented Programming (OOP)** are:

1. **Encapsulation** – Hiding data and controlling access through methods.
2. **Inheritance** – Creating a new class based on an existing class.
3. **Polymorphism** – Allowing the same method to behave differently in different classes.
4. **Abstraction** – Hiding implementation details and showing only essential features.

API

Q1: What is an API?

Answer:

An **API (Application Programming Interface)** is a way for different applications to communicate and exchange data with each other.

Keywords: Communication, Data Exchange, Applications

Q2: What is a REST API?

Answer:

A **REST API** is an API that follows REST principles and uses **HTTP methods** to allow communication between clients and servers.

Keywords: REST, HTTP, Client, Server

Q3: What are the common HTTP methods?

Answer:

The most common HTTP methods are **GET** to retrieve data, **POST** to create data, **PUT** to update data, and **DELETE** to remove data.

Keywords: GET, POST, PUT, DELETE

Q4: What is the difference between GET and POST?

Answer:

GET is used to retrieve data from the server, while **POST** is used to send new data to the server.

Keywords: Retrieve Data, Send Data

Q5: What is JSON?

Answer:

JSON (JavaScript Object Notation) is a lightweight format used to exchange data between applications and APIs.

Keywords: JSON, Data Format, API

Q6: What is an Endpoint?

Answer:

An **Endpoint** is a specific URL where an API can be accessed to perform a particular operation.

Keywords: URL, API Endpoint

Q7: What is the Client and Server?

Answer:

The **Client** sends requests, and the **Server** processes those requests and returns responses.

Keywords: Client, Server, Request, Response

Q8: What is an API Response?

Answer:

An API response is the data returned by the server after processing a request.

Keywords: Response, Data

Q9: What is Status Code 200?

Answer:

200 OK means the request was successful and the server returned the expected response.

Keywords: 200 OK, Success

Q10: What is Status Code 404?

Answer:

404 Not Found means the requested resource or endpoint does not exist.

Keywords: 404, Not Found

Q11: What is Status Code 500?

Answer:

500 Internal Server Error means there is a problem on the server side.

Keywords: 500, Server Error

Q12: Why are APIs important?

Answer:

APIs allow different systems and applications to communicate, integrate services, and share data efficiently.

Keywords: Integration, Communication, Data Sharing

سؤال مشهور جدًا

Q: Have you worked with APIs before?

لو معندكش خبرة كبيرة

I have basic experience working with APIs. I understand how APIs work, how requests and responses are exchanged, and how HTTP methods such as GET and POST are used. I am currently improving my practical experience.

Keywords: API, HTTP Methods, Requests, Responses